

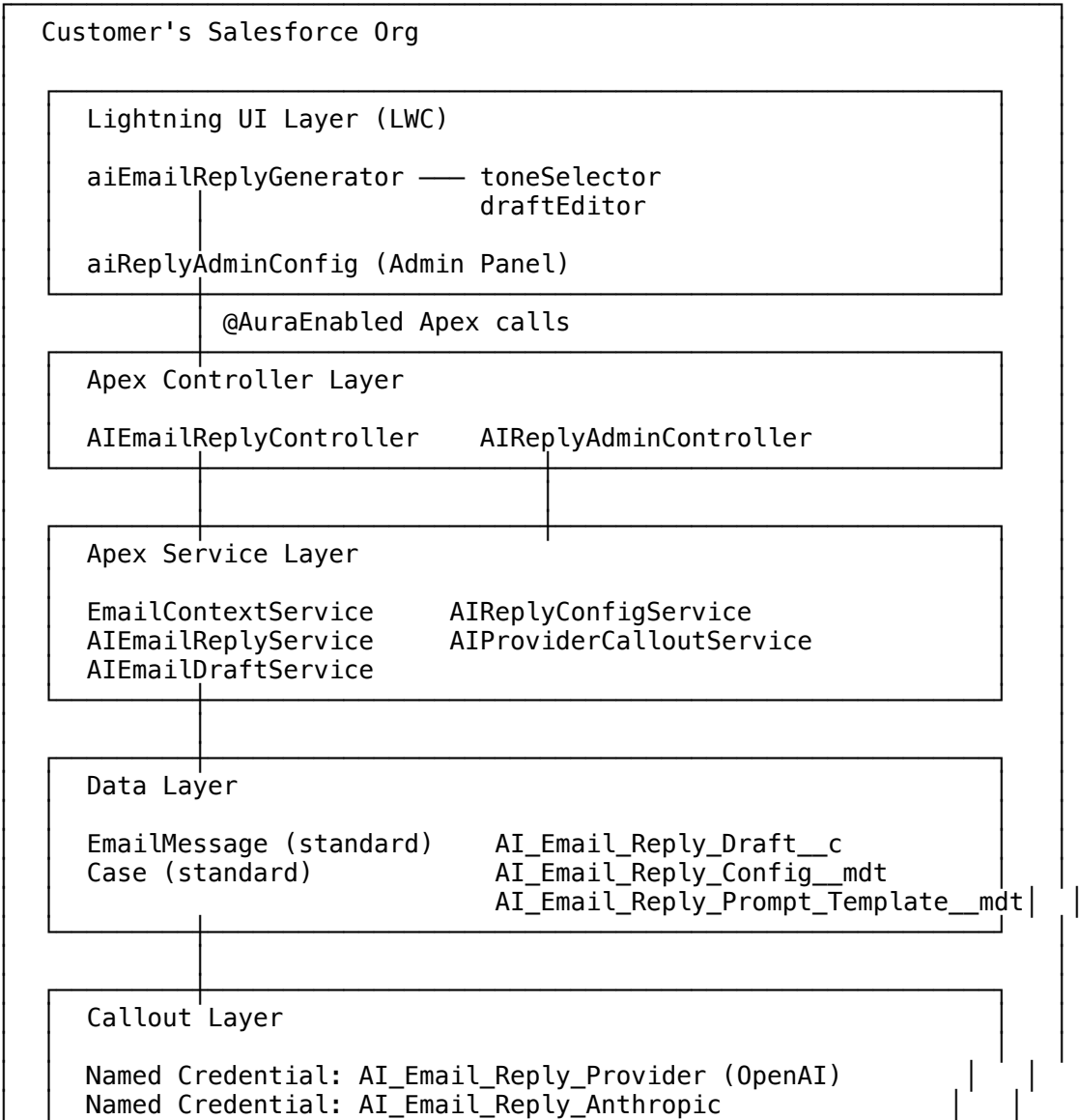
AI Email Reply Generator — Solution Architecture and Data Flow

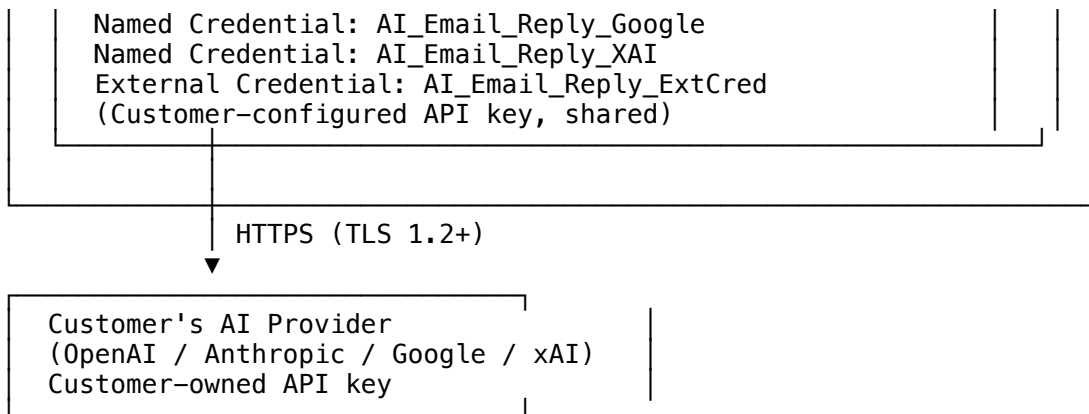
Architecture & Data Flow — AI Email Reply Generator

Company: MDS LLC Last Updated: May 7, 2026

1. High-Level Architecture

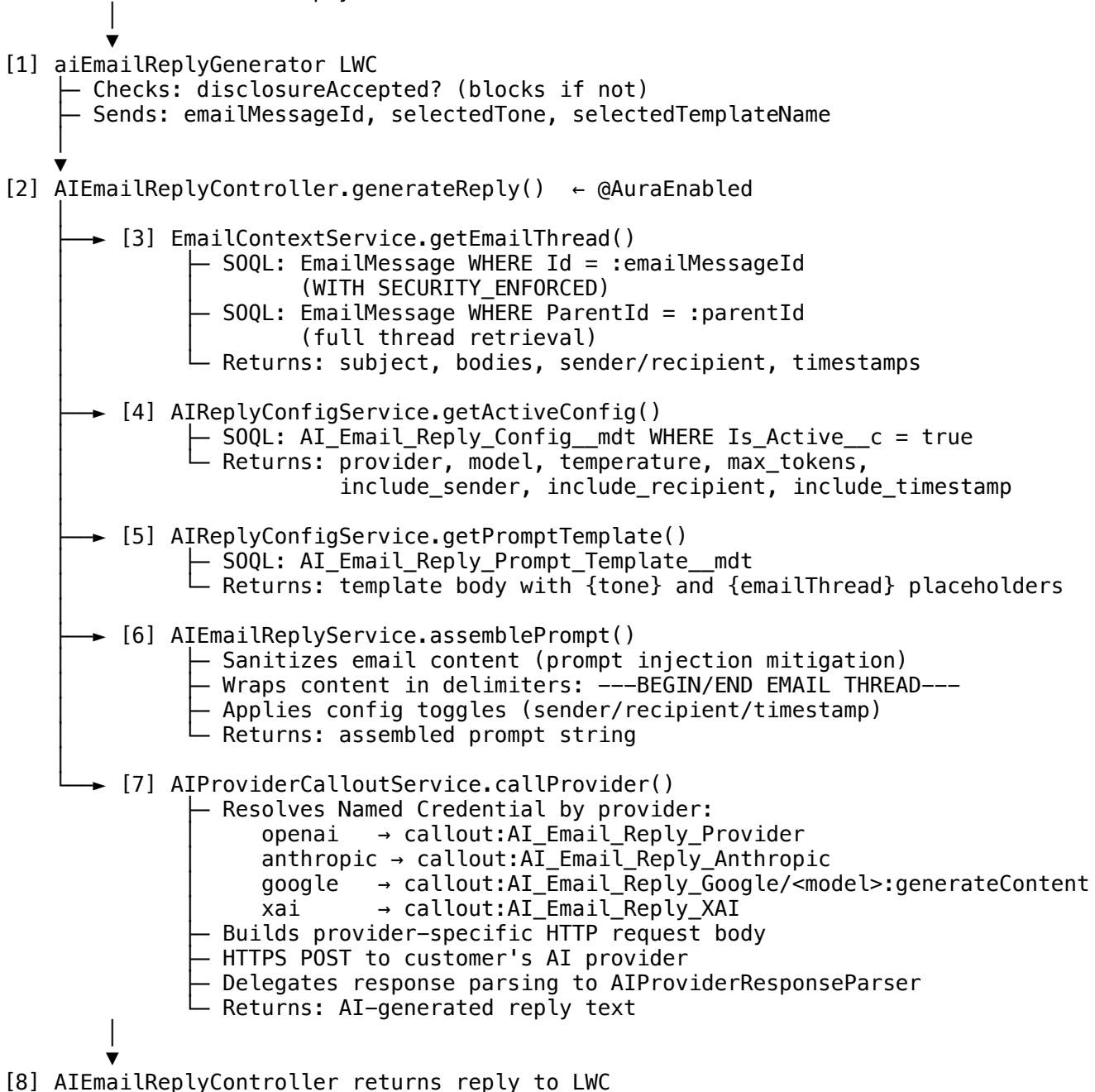
The AI Email Reply Generator is a 100% native Salesforce Lightning application. It consists of four LWC components, ten Apex classes, one custom object, two Custom Metadata Types, and four Named Credentials. No external servers or ISV-hosted infrastructure exist.





2. Data Flow — Reply Generation

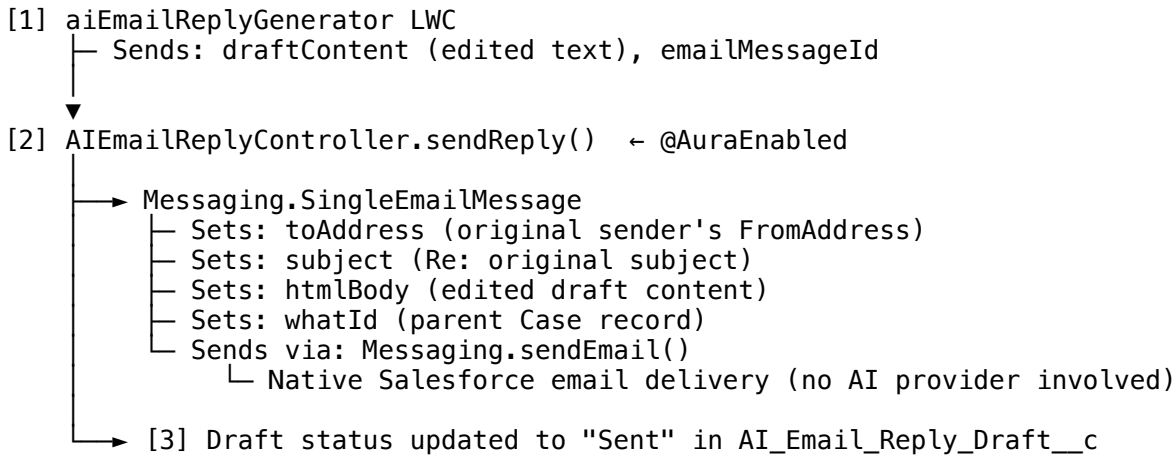
User clicks "Generate Reply"



[9] draftEditor displays generated text for review/editing

3. Data Flow — Email Sending

User clicks "Send Reply"

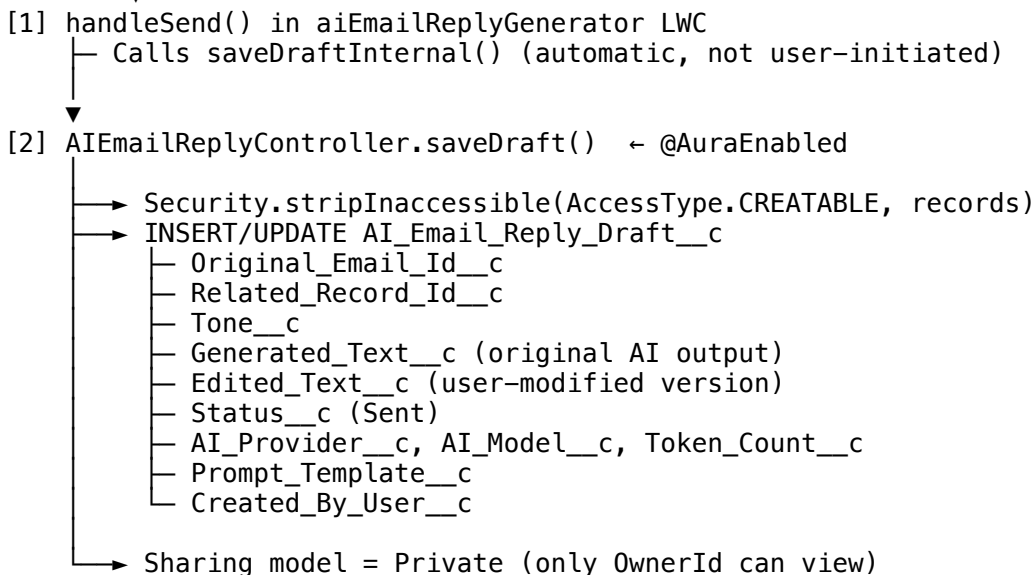


Key point: Email delivery uses Salesforce's native Messaging.sendEmail() API. No data passes through the AI provider at send time.

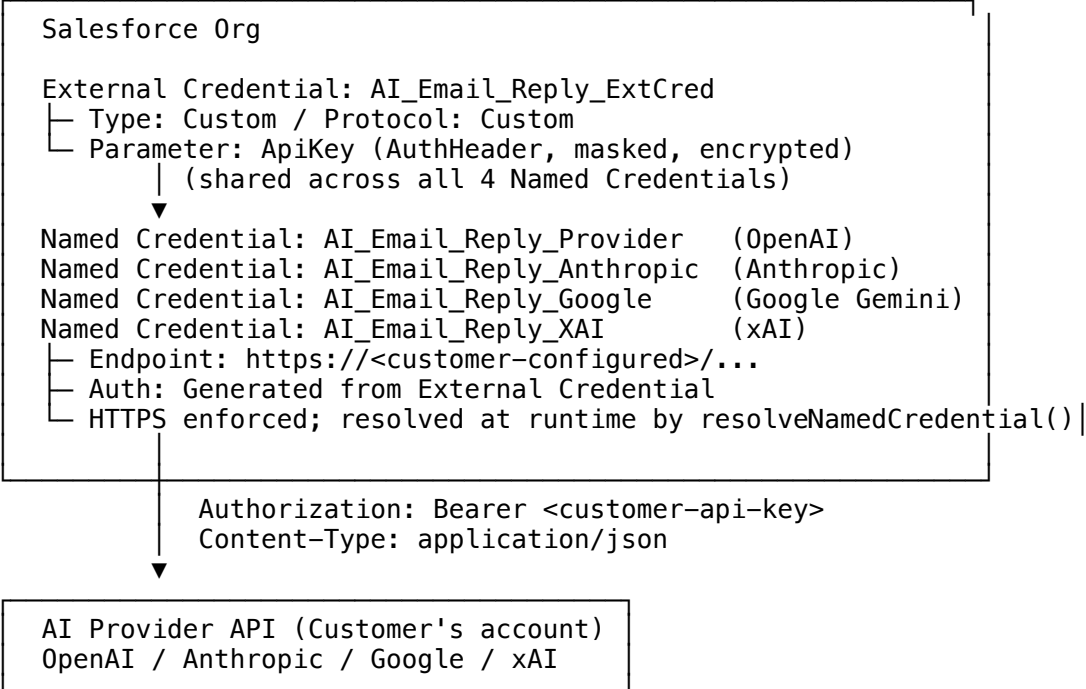
4. Data Flow — Draft Persistence (Internal)

Note: There is no user-facing "Save Draft" button. Draft records are created automatically when a reply is sent, to maintain an audit trail of AI-generated content.

User clicks "Send Reply"



5. Authentication & Credential Flow



Security controls:

- API key never appears in Apex code or debug logs.
- Named Credentials generate the Authorization header automatically via External Credential merge field.
- resolveNamedCredential() applies the managed namespace prefix (aiemailreply__) automatically when running in the packaged context.
- Customer configures both the endpoint URL and API key via Salesforce Setup UI.
- The AI_Email_Reply_ExtCred ships with placeholder value CONFIGURE_IN_SETUP.

6. Security Boundaries

Boundary	Description
Salesforce Org	All app logic, data storage, and UI run within the customer’s org
Named Credential	Single controlled exit point for external callouts (HTTPS only)
AI Provider	External — customer’s own account, customer’s own API key
MDS LLC	No data access — ISV code runs in customer’s org, no phone-home, no telemetry

7. Component Inventory

LWC Components

Component	Purpose	Placement
aiEmailReplyGenerator	Main orchestrator — email loading, generation, send	Case record pages, App Pages, Home Pages
toneSelector	Tone picker (Professional, Friendly, Empathetic, Formal, Concise, Custom)	Child of aiEmailReplyGenerator
draftEditor	Rich text editor for reviewing/editing AI drafts	Child of aiEmailReplyGenerator

Component	Purpose	Placement
aiReplyAdminConfig	Admin panel — config display, connection test, setup wizard	Dedicated Admin App Page

Apex Classes

Class	Layer	Purpose
AIEmailReplyController	Controller	@AuraEnabled methods for LWC (generate, send, get email)
AIReplyAdminController	Controller	@AuraEnabled methods for admin panel (config, test connection)
EmailContextService	Service	Queries EmailMessage records and assembles email thread context
AIReplyConfigService	Service	Reads CMTD configuration and prompt templates
AIEmailReplyService	Service	Assembles prompts, sanitizes input, orchestrates generation
AIProviderCalloutService	Service	Performs HTTP callout via Named Credential; delegates parsing to parser
AIProviderResponseParser	Service	Parses HTTP responses from all 4 provider formats into standard objects
AIEmailDraftService	Service	CRUD operations on AI_Email_Reply_Draft__c
AIReplyLogger	Utility	Centralized logging — logs only e.getMessage() at LoggingLevel.ERROR
PostInstallHandler	Installer	Post-install script — creates unmanaged AI_Email_Reply_API_Access permission set and assigns it to installing user

Data Objects

Object	Type	Sharing
AI_Email_Reply_Draft__c	Custom Object	Private
AI_Email_Reply_Config__mdt	Custom Metadata Type	—
AI_Email_Reply_Prompt_Template__mdt	Custom Metadata Type	—

Credentials

Name	Type	Provider
AI_Email_Reply_Provider	Named Credential	OpenAI (default)
AI_Email_Reply_Anthropic	Named Credential	Anthropic
AI_Email_Reply_Google	Named Credential	Google Gemini
AI_Email_Reply_XAI	Named Credential	xAI (Grok)
AI_Email_Reply_ExtCred	External Credential (Custom auth)	All providers

Post-Install Behavior

On package installation, PostInstallHandler runs automatically and:

1. Creates an unmanaged Permission Set named AI_Email_Reply_API_Access (if it does not already exist).
2. Assigns it to the installing user.
3. The admin must then manually add External Credential principal access to this Permission Set via Salesforce Setup → External Credential Principal Access — as guided by the in-app Setup Wizard.